

MUHAMMAD YUSUF JAAFAR

EXPERIENCED RISK-BASED INSPECTION (RBI) ENGINEER

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EXECUTIVE SUMMARY

Highly accomplished Risk-Based Inspection (RBI) Engineer with over 7 years of hands-on experience in asset integrity management for critical equipment and piping. Holding both a Bachelor's in Mechanical Engineering and a Master's in Petroleum Engineering, I specialize in project planning, corrosion rate analysis, remaining life calculations, and risk assessments in compliance with API 510, API 570, API 580/581, and DOSH standards. Adept at developing and executing inspection programs, leveraging NDT techniques to enhance safety, reliability, and operational efficiency. Proven track record in managing Pressure Relief Device (PRD) inspections and ensuring regulatory compliance. Strong leadership and problem-solving skills, with a commitment to achieving results in the oil and gas industry.

WORK EXPERIENCE

1) RBI Engineer | BUMITECH GLOBAL COMPANY | Labuan, Malaysia

Sep 2023 – Present

Manage Risk-Based Inspection for PETRONAS Chemical Methanol (Plant 2) as main client, improving the safety, reliability, and cost-efficiency of industrial operations by strategically focusing inspection resources delivering exceptional services and leading cross-functional teams to achieve company goals.

- **Comprehensive Risk Assessments and Compliance:** Led in-depth risk assessments for critical DOSH and non-DOSH equipment, ensuring that all assets complied with stringent safety regulations including identified high-risk areas and mitigated potential hazards, significantly enhancing plant safety and operational reliability.
- **Inspection Program Development and Execution:** Spearheaded the creation and execution of robust inspection programs, encompassing detailed inspection schedules, non-destructive testing (NDT), and RBI methodologies to minimize downtime and optimizing asset performance.
- **Corrosion Rate and Remaining Life Analysis:** Delivered expert corrosion rate (CR) and remaining life (RL) calculations for critical equipment and piping while extended the lifespan of high-value assets by predicting potential failures and implementing timely interventions, aligning with DOSH and international safety standards.
- **Asset Integrity and Data Management:** Manage the integrity of Asset Performance Management systems by thoroughly managing data for piping and equipment including enhanced decision-making processes, improving asset reliability and operational efficiency.
- **Damage Mechanism and Failure Mode Assignments:** Develop proactive maintenance strategies and accurately assigned damage mechanisms to equipment components in line with API 580/581 standards considering failure modes, general corrosion rates, and expected damage patterns.
- **Regulatory Compliance and Certificate Issuance:** Manage timely issuance of Certificates of Fitness (CF) using Ultrasonic Thickness Testing Measurements (UTTM) data to ensure compliance with regulatory requirements, facilitating seamless compliance with DOSH regulations, preventing costly delays.
- **Historical Data Review and Critical Analysis:** Become the focal points in reviewing static equipment and piping records, including design, inspection, maintenance, and operational data including provided valuable insights that contributed to the continual improvement of plant operations.

Key Achievements:

- **Enhanced Inspection Accuracy:** Successfully updated and optimized PRBI V3 data, improving the precision of inspection and maintenance planning, which resulted in a 15% reduction in unplanned equipment downtime.
- **Regulatory Compliance Excellence:** Played a pivotal role in ensuring full compliance with DOSH and international standards, leading to the seamless issuance of CFs for critical equipment and enhancing overall plant safety and operational continuity.

Related Skills: Project Planning | Problem-Solving | Data Analysis | Stakeholders Engagement | Asset Integrity

2) **Plant Inspector (Resident) | LOTT INSPECTION COMPANY | Labuan, Malaysia** **Sep 2019 – Aug 2023**

- **Inspection Planning and Execution:** Leveraged expertise in API 510 and API 570 standards to ensure compliance with design, service, and jurisdictional requirements across refinery, petrochemical, and shutdown operations as well as developing and executing comprehensive on-stream and shutdown inspection plans.
- **Integrity Assessments and Corrosion Management:** Identified critical areas prone to deterioration, such as dead legs, injection points, and buried piping to address corrosion under insulation (CUI), erosion, and corrosion issues, ensuring long-term plant integrity.
- **NDT and Welding Procedure Reviews:** Validated welding procedures (WPS, PQR, WQT) and NDT techniques (Radiography, Dye Penetrant, Magnetic Particle, Ultrasonic Testing) to ensure compliance with industry codes, standards, and specifications as well as provided recommendations on repair methods and materials selection.
- **Thorough Inspection of Critical Equipment:** Conducted thorough visual inspections and complemented them with non-destructive testing (NDT) on critical equipment, such as pressure vessels, heat exchangers, pipelines, and storage tanks.
- **New Installation and Modification Oversight:** Leading team in new equipment installations and modification of existing equipment on piping projects (tie-ins, hot taps, etc.) especially those involving pressurized parts, to ensure compliance with DOSH and API 510 standards Inspected
- **Pressure Relief Devices (PRDs) Management:** Conducted regular failure rate analyses and witnessed testing procedures (pop test, leak test, back pressure test), ensuring the integrity of pressure relief systems as well as maintained a comprehensive list of all safety devices, such as PRVs, PSVs, and rupture discs.

3) **Freelance Inspector – Shutdown/Turnaround Projects | Terengganu, Malaysia** **Jan 2017 – Jun 2022**

Performed comprehensive inspections and analyses of static equipment and piping systems during shutdowns, ensuring mechanical integrity, documenting findings, and verifying repair effectiveness while adhering to safety protocols. Key projects included turnarounds and inspections for major clients such as PETRONAS, LOTTE Chemical Titan, and BASF PETRONAS Chemicals, covering various plants and modules.

4) **Engineering Intern | PRESTON SHIPYARD COMPANY | Labuan, Malaysia** **Jul 2013 – Sept 2013**

Assisted seniors in vessel inspection, maintenance, and repair, supported engineers in project planning, conducted ship structure inspections, and contributed to technical reports, gaining hands-on experience in welding, fabrication, and corrosion control, while ensuring compliance with safety standards. Key projects included the total refit of KD GANAS (Navy Vessel) and KM MABUL (Maritime Vessel).

EDUCATION BACKGROUND

Universiti Teknologi Malaysia (UTM) | Master of Petroleum Engineering **2016**

Related Project: Low Salinity Water Flooding (LSWF) Simulation Study

- To study the mechanism that contribute to the increasing oil recovery with LSWF technique using ECLIPSE 100 black oil simulator and manage to prove that wettability alteration significantly contributes to increased recovery.

Universiti Tun Hussein Onn Malaysia (UTHM) | Bachelor of Mechanical Engineering **2014**

Related Project: The Effect of Composition and Sintering Temperature of Silica (SiO₂) and Claytan's Raw Materials

- To study on the effect of composition and sintering temperature of silica (SiO₂) and CLAYTAN's raw materials, and successfully reduced the sintering temperature by 15% while maintaining the physical and mechanical properties of the products

ACHIEVEMENT AND ADDITIONAL INFORMATION

Major Projects Shutdown/turnaround involved:

1. **PETRONAS Chemical Methanol (Plant 2)**
 - a. Affinity Test Services SDN BHD: **Piping Inspector** - Plant Turnaround | May 2022
2. **LOTTE Chemical Titan SDN BHD (Plant PP3)**
 - a. MJ Inspection SDN BHD: **Static Equipment Inspector** - Plant Turnaround | October 2019
3. **Malaysia LNG**
 - a. Lott Inspection SDN BHD: **Static Equipment Inspector** - Modul 2 Pit Stop | August 2019
 - b. Lott Inspection SDN BHD: **API 510 Inspector** - Plant LPG SSI - RBI | August 2019
 - c. Lott Inspection SDN BHD: **Static Equipment Inspector** - Modul 7 Plant Turnaround | June 2019
4. **BASF PETRONAS Chemicals Sdn. Bhd.**
 - a. HRSB Holdings SDN BHD: **QC Inspector** - HR-PIB Plant Turnaround | April 2019
 - b. HRSB Holdings SDN BHD: **QC Inspector** - Acrylic Plant Turnaround | February 2019
5. **PETRONAS Gas Berhad**
 - a. NDE Consultancy Services SDN BHD: **Static Equipment Inspector** - T.Sulong Export Terminal Plant Turnaround | July 2018
 - b. NDE Consultancy Services SDN BHD: **Static Equipment Inspector** - GPP 2 Plant Turnaround | February 2017
6. **PETRONAS Chemical MTBE PDH SDN BHD**
 - a. NDE Consultancy Services SDN BHD: **Static Equipment Inspector** - Plant Turnaround | April 2018
7. **PETRONAS Chemical Olefins, Glycols & Derivatives SDN BHD**
 - a. NDE Consultancy Services SDN BHD: **Static Equipment Inspector** - Plant Turnaround | July 2017

Certification:

- API 580 | Risk Based Inspection - Certification No: 100448, Expiration Date: June 30, 2027
- API 577 | Welding Inspection and Metallurgy - Certification No: 99090, Expiration Date: Feb 28, 2027
- API 570 | Piping Inspector - Certification No: 95641, Expiration Date: Mar 31, 2026
- API 510 | Pressure Vessel Inspector - Certification No: 76342, Expiration Date: Sept 30, 2026
- Oil and Gas Safety Passport (OGSP-R) - NIOSH, No: OG003177N, Valid Thru: 19/05/2026
- Authorized Entrant and Standby Person for Confined Space - NIOSH, No: NW-SBRO-AE-R-2114-V, Expiration Date: 09 October 2025
- PTW, JHA & LOTO Training for Contractor as Receiving Authority - PETRONAS Chemical Methanol SDN BHD, Date: 27 - 29 October 2021
- API 577 Welding Inspection and Metallurgy Professional Course - INSTECH Consulting, Date: 26 – 27 Sept 2020
- Authorized Work Leader (AWL) - PETRONAS Chemical Methanol SDN BHD, Date: 21 - 22 January 2020
- API 510 Pressure Vessel Inspector Certification Course - INSTECH Consulting, Date: 18 – 19 Feb 2017

Professional Affiliation:

- Board of Engineering Malaysia (BEM), Graduate Member Ref: 98768A
- The Institute of Engineer Malaysia (IEM), Graduate Member Ref: 72416

Languages: Fluent in English and Malay Language

Hard-Skills: Microsoft Office | Reservoir Simulation | Engineering Design | Computational Fluid Dynamics | Programming Language | Failure Mode Analysis | Metallography and Thermography | Asset Integrity Management | Shutdown and Turnaround Inspection

Strengths: Willing to learn | Team Player | Flexible | Good time management | Work independently | Analytical | Detailed-Oriented